Expanded saved-record tables

These unchanged numerical tables are an optional reproduction companion, not extra pages of the main paper. They preserve the frozen records, hypotheses and interval definitions. Figure 3 refers to the main manuscript; its OSLD prespecified rows retain their original scope.

Table 1: Complete exploratory hook/control contrasts (pp). Basic 95% intervals from 10,000 shared query-identity cluster draws; conditional on trained models and fixed gallery, unadjusted and not simultaneous. No confirmatory p-values. R1: Recall@1; Loc: query-logo localization. Seeds list paired differences.

Contrast	Metric	Estimate	95% interval	Seeds 0/1/2
BGER-F − BGER-H	Loc	+0.3	[−0.1, +0.7]	+0.9/−0.6/+0.6
BGER-F − CE	Loc	−1.2	[−2.0, −0.3]	−0.2/−2.5/−0.7
BGER-F − Aux-BCE	Loc	−3.0	[−4.0, −1.9]	−3.2/−4.1/−1.6
BGER-F − CAM-BCE	Loc	−15.2	[−17.5, −12.6]	−13.3/−17.1/−15.0
BGER-F − ABN-box	Loc	−1.7	[−2.5, −0.8]	−1.9/−2.7/−0.4
Gate-only − CE	R1	−0.8	[−1.3, −0.2]	−0.3/−1.4/−0.6
Gate-only − CE	Loc	+0.5	[−0.3, +1.2]	+1.3/−0.3/+0.4
BGER-H − gate-only	R1	+3.3	[+2.1, +4.4]	+3.4/+4.2/+2.3
BGER-H − gate-only	Loc	−1.9	[−2.8, −1.0]	−2.4/−1.5/−1.8
Shifted-box − CE	R1	−0.6	[−1.2, +0.1]	−0.3/−0.2/−1.1
Shifted-box − CE	Loc	−0.4	[−1.1, +0.3]	−0.2/−0.9/0.0
BGER-H − shifted-box	R1	+3.1	[+1.9, +4.2]	+3.4/+3.1/+2.8
BGER-H − shifted-box	Loc	−1.1	[−1.9, −0.2]	−0.9/−1.0/−1.4
Gate-only − Aux-BCE	R1	−0.6	[−1.2, 0.0]	−0.7/−1.1/+0.1
Gate-only − Aux-BCE	Loc	−1.4	[−2.3, −0.4]	−1.7/−1.9/−0.5
Shifted-box − gate-only	R1	+0.2	[−0.4, +0.7]	0.0/+1.1/−0.6
Shifted-box − gate-only	Loc	−0.8	[−1.6, −0.1]	−1.5/−0.6/−0.4

Table 2: All paired contrasts of the OSLD test set (percentage points; estimate and basic 95% interval from the shared draws; descriptive except BGER−CE R@1 and $\bar{L}^{q}$, the prespecified hypotheses shown in Figure 3).

Contrast	R@1	R@10	mAP	$\bar{L}^{q}$	Jq@1	EIB	Sec. $\bar{L}^{q}$
BGER−CE	+2.5 [+1.4, +3.6]	+6.4 [+4.9, +7.9]	+3.0 [+2.1, +3.8]	−1.5 [−2.2, −0.6]	−0.5 [−0.9, −0.1]	+0.3 [−0.2, +0.9]	+16.4 [+14.2, +18.5]
BGER−Aux-BCE	+2.7 [+1.7, +3.6]	+6.3 [+4.4, +8.0]	+3.0 [+2.1, +3.8]	−3.3 [−4.3, −2.2]	−0.5 [−0.9, −0.1]	−1.1 [−1.7, −0.4]	+14.4 [+12.3, +16.4]
BGER−CAM-BCE	+1.5 [+0.5, +2.5]	+3.0 [+1.3, +4.6]	+1.6 [+0.9, +2.4]	−15.5 [−17.9, −12.9]	−2.5 [−3.5, −1.5]	−8.1 [−9.3, −6.8]	−10.5 [−13.1, −7.9]
BGER−ABN-box	+2.6 [+1.5, +3.7]	+4.5 [+2.8, +6.2]	+2.6 [+1.8, +3.3]	−2.0 [−2.8, −1.0]	−0.3 [−0.6, +0.1]	−1.6 [−2.1, −1.0]	+14.5 [+12.4, +16.4]
Aux-BCE−CE	−0.2 [−0.9, +0.6]	+0.2 [−0.9, +1.3]	0.0 [−0.4, +0.4]	+1.8 [+0.9, +2.7]	+0.1 [−0.3, +0.4]	+1.4 [+1.0, +1.8]	+2.0 [+0.8, +3.1]
CAM-BCE−CE	+1.1 [+0.2, +1.9]	+3.5 [+2.2, +4.7]	+1.4 [+0.7, +2.0]	+14.0 [+11.7, +16.2]	+2.0 [+1.1, +2.8]	+8.5 [+7.4, +9.4]	+26.9 [+23.5, +30.1]
ABN-box−CE	−0.1 [−0.9, +0.7]	+1.9 [+0.6, +3.2]	+0.4 [−0.2, +1.0]	+0.5 [−0.2, +1.2]	−0.2 [−0.6, +0.2]	+1.9 [+1.5, +2.4]	+1.9 [+0.7, +3.1]

Table 3: Complete OSLD development record (official validation: 1,295 queries, 7,272 train+validation templates; %). Seed-0 rows select λ within each arm (Recall@1 within 1 pp of the arm's best, then highest $\bar{L}^{q}$, ties to the smaller λ); seeds 1–2 lock thresholds.

Arm	Seed	λ	R@1	R@10	mAP	$\bar{L}^{q}$	Jq@1	t	Sel.
CE	0	0	5.8	15.1	6.4	3.9	1.7	0.55	✓
Aux-BCE	0	0.5	5.9	16.4	6.7	5.7	1.8	0.60	✓
Aux-BCE	0	2	4.2	15.7	5.9	5.7	0.5	0.65	
BGER	0	0.5	10.4	25.5	11.2	2.4	0.6	0.55	✓
BGER	0	2	9.3	22.5	9.8	3.9	0.7	0.50	
CAM-BCE	0	0.5	8.8	21.4	9.5	17.4	5.1	0.50	✓
CAM-BCE	0	2	2.7	9.0	3.2	6.9	1.1	0.60	
ABN-box	0	0.5	4.2	14.8	5.9	4.2	0.8	0.70	
ABN-box	0	2	5.6	20.3	7.3	4.2	0.7	0.65	✓
CE	1	0	4.8	14.1	5.7	3.9	0.7	0.65	✓
Aux-BCE	1	0.5	6.9	17.1	7.0	4.7	1.1	0.65	✓
BGER	1	0.5	10.3	23.1	10.6	2.2	0.8	0.50	✓
CAM-BCE	1	0.5	9.3	21.9	9.6	16.6	5.2	0.55	✓
ABN-box	1	2	7.3	20.4	8.1	4.3	1.4	0.65	✓
CE	2	0	5.9	15.6	6.8	4.1	1.2	0.60	✓
Aux-BCE	2	0.5	4.6	15.8	6.1	3.6	0.8	0.60	✓
BGER	2	0.5	9.2	22.8	9.8	2.6	0.8	0.55	✓
CAM-BCE	2	0.5	8.0	20.2	8.7	20.0	3.5	0.60	✓
ABN-box	2	2	6.0	18.5	7.6	3.3	0.8	0.70	✓

Table 4: Fixed-candidate diagnostic (query side; seed-averaged; same CE-seed-0 d^{+} and d^{-} for all arms). *Change between candidates*: raw magnitude (median $|\ln|$ ratio of the maps' maximum absolute values), normalized shape (L1 distance of ReLU-normalized maps, 0–2; mean defined correlation, undefined count) and the thresholded verdict (rate of differing localization verdicts, %, with the bounds implied by the two hit rates). *Localization*: hit rate for d^{+}/d^{-} and a centre prior evaluated at each arm's locked threshold (%; not tuned for the prior). No single column ranks overall candidate dependence; descriptive, not a faithfulness measure.

	Change between candidates				Localization	
Arm	Raw	L1	Corr. (undef.)	Verdict [bounds]	d^{+}/d^{-}	Prior
Primary (Grad-CAM)						
CE	0.43	1.25	0.19 (0)	7.9 [4.6, 10.2]	7.4/2.8	4.0
Aux-BCE	0.44	1.25	0.19 (0)	10.0 [5.1, 12.7]	8.9/3.8	4.3
BGER	0.42	1.22	0.25 (0)	4.4 [3.4, 4.8]	4.1/0.7	3.1
CAM-BCE	0.51	1.16	0.33 (0)	18.8 [10.6, 43.5]	27.0/16.4	3.3
ABN-box	0.45	1.31	0.13 (0)	7.6 [4.5, 9.3]	6.9/2.4	5.1
Secondary (decomposition)						
CE	0.21	0.12	0.92 (0)	9.9 [2.0, 24.1]	13.0/11.1	5.6
Aux-BCE	0.17	0.12	0.92 (0)	10.3 [3.5, 27.7]	15.6/12.1	5.6
BGER	0.16	0.08	0.99 (0)	10.1 [1.3, 54.8]	28.1/26.8	5.8
CAM-BCE	0.17	0.12	0.97 (0)	8.3 [3.4, 79.1]	41.3/37.8	6.0
ABN-box	0.18	0.14	0.92 (0)	10.3 [2.7, 26.0]	14.3/11.6	4.7

Related CUB classification records

The comparison below retains the differences in task, maps, metrics, pretraining, selection and supervision. It is related evidence, not a controlled transfer result.

Table 5: The CUB study is related evidence, not a controlled transfer: these differences between the two studies all vary at once.

	CUB-200-2011	OSLD
Task	closed-set, 200 classes	open-set retrieval of unseen identities
Evaluated map	predicted-class Grad-CAM of a logit	Grad-CAM of the cosine to the top-1 template
Localization	MaxBoxAccV2 envelope (101 thresholds, best per IoU)	IoU $\geq .5$ at a validation-locked threshold
Objects	birds, mostly large boxes	logos, median box 3.9% of the image
Pretraining	ImageNet V2 weights	ImageNet V1 weights
Checkpoint	best validation accuracy	final epoch
Box target, loss	$\geq$1 cell; per-image balance, clipped prior	any-overlap cells; batch-level balance
Weights	$\lambda = 2$, three seeds (CAM-BCE also .5, post hoc)	$\lambda \in \{.5, 2\}$ selected on validation
Extra arm	Residual-BCE $(1 + 0.5M) \odot H$	ABN-box

Table 6: Complete existing seed-0 coefficient grid of the matched generation (full CUB test). Observed models only; no further weights were trained.

Arm	λ	Acc.	V2	EIB	PG
CE	—	0.7415	0.5970	0.7883	0.9670
Aux-BCE	0.125	0.7430	0.6158	0.7934	0.9724
Aux-BCE	0.5	0.7416	0.6520	0.7999	0.9796
Aux-BCE	2	0.7375	0.7030	0.8106	0.9902
Residual-BCE	0.125	0.7530	0.6467	0.8065	0.9812
Residual-BCE	0.5	0.7537	0.6822	0.8112	0.9812
Residual-BCE	2	0.7423	0.7113	0.8140	0.9874
BGER	0.125	0.7427	0.6401	0.8242	0.9865
BGER	0.5	0.7489	0.6846	0.8374	0.9914
BGER	2	0.7489	0.7190	0.8406	0.9950
CAM-BCE	0.125	0.7440	0.6610	0.8457	0.9933
CAM-BCE	0.5	0.7451	0.7207	0.8561	0.9946
CAM-BCE	2	0.7356	0.7784	0.8251	0.9969

Table 7: All three-seed runs (full CUB test). With Table 6 this covers all 25 saved runs of the generation.

Method	λ	Seed	Acc.	V2	EIB	PG
CE	—	0	0.7415	0.5970	0.7883	0.9670
CE	—	1	0.7547	0.5853	0.7870	0.9636
CE	—	2	0.7523	0.5881	0.7862	0.9655
Aux-BCE	2	0	0.7375	0.7030	0.8106	0.9902
Aux-BCE	2	1	0.7513	0.6892	0.8070	0.9874
Aux-BCE	2	2	0.7534	0.6915	0.8032	0.9883
Residual-BCE	2	0	0.7423	0.7113	0.8140	0.9874
Residual-BCE	2	1	0.7554	0.7084	0.8161	0.9884
Residual-BCE	2	2	0.7532	0.7102	0.8133	0.9893
BGER	2	0	0.7489	0.7190	0.8406	0.9950
BGER	2	1	0.7497	0.7119	0.8401	0.9936
BGER	2	2	0.7466	0.7211	0.8399	0.9946
CAM-BCE	0.5	0	0.7451	0.7207	0.8561	0.9946
CAM-BCE	0.5	1	0.7530	0.7128	0.8552	0.9945
CAM-BCE	0.5	2	0.7432	0.7014	0.8542	0.9952
CAM-BCE	2	0	0.7356	0.7784	0.8251	0.9969
CAM-BCE	2	1	0.7423	0.7625	0.8415	0.9960
CAM-BCE	2	2	0.7466	0.7684	0.8306	0.9960